

v4.5.5_review_summary_2026-05-04

v4.5.5 Review Summary — 2026-05-04 (Four Rounds)

Final deck: docs/menopause_sns_gtex_enhanced_2026-05-04_v4.5.5.pptx (50 slides) **Final explainer:** docs/menopause_sns_gtex_enhanced_2026-05-04_v4.5.5_explainer.md (PDF rendered) **Audit + expansion windows:** 2026-04-30 → 2026-05-04 **Predecessors audited:** v4.5.1 (2026-04-22) → v4.5.2 (2026-05-01) → v4.5.3 (2026-05-01) → v4.5.4 (2026-05-03) → v4.5.5 (2026-05-04)

The work proceeded in four rounds. Rounds 1 and 2 were correctness-focused audits on the 37-slide v4.5.3 deck (14 fixes spanning prose, attribution, and documentation; no fabricated data). Round 3 (2026-05-03) was an expansion: 13 new analysis slides triggered by collaborator-meeting feedback. Round 4 (2026-05-04) was an adversarial audit of the v4.5.4 expansion that identified **9 issues including 2 HIGH-severity factual errors (gene counts) and 1 HIGH-severity statistical overstatement**. v4.5.5 corrects all 9. This is the consolidated four-round report.

1. Reviewer panel and new contributors

Round	#	Role	Lens	Items checked	Issues found
1	R1	Stats auditor	Every number vs source TSV	76	0
1	R2	Methods auditor	Every model vs R script	23	0 (2 LOW doc gaps)
1	R3	Interpretation/ concordance auditor	Direction, hedging, concordance, attributions	full doc	1 HIGH + 3 MEDIUM + 2 LOW
2	R4	Slide-content auditor	Every slide text vs explainer prose, slide-by- slide	37 slides	5 MEDIUM + 4 LOW
2	R5	Figure auditor	Every figure path exists; PPTX images by MD5	18 unique PNGs + 32 PPTX images	0 missing (9 LOW path-stem mismatches in explainer)
2	R6	Deep numerical auditor	22 uncovered + 4 newly- introduced gene-direction counts	22	0
2	R7	Patch validator	Round-1 fixes correctly applied, no regressions	6 fixes + 37 footers	0
3	R8	R analysis (expansion)	New ADRB3 mirrors, sensitivity stratification, signature scoring	New script + 8 new TSVs + 13 new PNGs	—

Round	#	Role	Lens	Items checked	Issues found
3	R9	PPTX patch (expansion)	13-slide insertion, footer renormalization	13 inserts + 50 footers	0 ordering errors
3	R10	Documentation (expansion)	Explainer + review summary updated to v4.5.4	Predecessor doc + slide-number map	—
4	R11	Stats auditor (v4.5.4)	Every v4.5.4-introduced number vs source TSV	41	3 MISMATCH (2 HIGH gene counts, 1 MED tertile %)
4	R12	R-script methods auditor (v4.5.4)	New v4.5.4 R script vs claimed methods	26	0 mismatches (R logic fully correct)
4	R13	Slide content + figure auditor (v4.5.4)	Every new slide vs explainer; figures by MD5	13 slides + 13 PNGs	4 MED cross-ref + 1 MED stale ref + 1 LOW R ² + 1 LOW title
4	R14	Interpretation / claim-strength auditor (v4.5.4)	Three headline findings + lipolysis personal notes	4 claims	1 HIGH overreach (Claim 1) + 1 MED partial (Claim 2) + 1 MED partial (Claim 3) + 1 LOW partial (Claim 4)

2. Consolidated coverage statistics

Audit dimension	Items checked	Verified	Mismatches	New in Round 3
Numerical claims (R1 + R6)	98 (76 + 22)	98	0	Not re-audited (no v4.5.3 numbers changed in v4.5.4)
Method specifications (R2)	23	23	0	New R script (R8) follows the same conventions as the R2-audited 2026-04-21 script
Slide-by-slide content (R4)	37 v4.5.3 slides	37 fully clean post-Round-2	—	13 new v4.5.4 slides (not yet adversarially audited; built fresh and self-consistent)
Figures (R5)	18 + 32 PPTX	18 + 32	0 missing	13 new PNGs all generated and embedded
v4.5.x patch validity (R7)	6 fixes + 37 footers	6 + 37	0 regressions	13 new slides + 50 footers verified by R9 (orientation + footer sweep)
Interpretation/ concordance (R3)	full document	—	—	New v4.5.4 prose is self-consistent; not yet adversarially audited

Net through end of Round 2: 0 numerical or methodological errors; 14 prose/attribution/documentation issues all resolved. **Net new in Round 3:** 13 expansion slides backed by 8 new tables + 13 new figures; significance-marker convention applied to all new figures.

3. Round 3 — what was added (full ledger)

3.1 New R analysis script

R/next_steps_2026-05-03_v4.5.4_adrb3_signature_stratification.R (≈700 lines, runs in ~2 minutes after VST re-extraction).

Inputs:

- tables/next_steps_2026-04-21_adrb2_age_stratified/scores_with_extra_genes_both_depots.tsv (existing — has ADRB1/2/3 + cell fractions + 18 gene-set scores)
- references/menopause_signature_senMayoLike_v2.tsv (30 genes, directional)
- references/menopause_signature_curated.tsv (70 genes, no direction)
- GTExDataV10/gene_reads_v10_adipose_{subcutaneous,visceral_omentum}.gct.gz (re-VST for the signature genes that aren't in the cached TSV)

New tables produced (in tables/next_steps_2026-05-03_v4.5.4/):

File	Schema	Rows
adrb2_adrb3_age_stratified_female_correlations.tsv	Combined long-format correlations: receptor × stratification × depot × target × age_stratum	1,302
adrb2_adrb3_age_correlation_fisherz_3strats.tsv	Fisher-z comparisons across the 3 stratifications, both receptors	138
adrb3_age_correlation_young_vs_older_fisherz.tsv	ADRB3-only slice of the above	78
sensitivity_50s_excluded_fisherz.tsv	broad_50s_excluded slice, both receptors	68
saltiel_style_regression_stats_adrb3.tsv	16 rows mirroring Slide 33 for ADRB3	17
menopause_score_per_sample.tsv	Per-sample senMayoLike v2 + curated scores	1,302
score_tertile_vs_age_bin_concordance.tsv	Tertile vs menopause_group cross-tabulation	39
score_tertile_correlations_both_receptors.tsv	Per-tertile correlation results	168

New figures (in figs/next_steps_2026-05-03_v4.5.4/):

File	Slide	Purpose
adrb3_correlation_panel.png	7	ADRB3 vs gene-set scores (8 panels, mirror of Slide 6 left)
adrb3_vs_cibersort_fractions.png	7	ADRB3 vs CIBERSORT fractions (mirror of Slide 6 right)
adrb3_age_stratified_forest_female.png	28	ADRB3 forest, 20-49 vs 50-79

File	Slide	Purpose
sensitivity_50s_excluded_forest_female.png	29	Sensitivity comparison: 50s incl vs excl
saltiel_adrb3_scatters_female_subcutaneous.png	35	Saltiel ADRB3 female subq
saltiel_adrb3_scatters_female_visceral.png	37	Saltiel ADRB3 female visc
saltiel_adrb3_scatters_male_subcutaneous.png	39	Saltiel ADRB3 male subq
saltiel_adrb3_scatters_male_visceral.png	41	Saltiel ADRB3 male visc
menopause_score_distribution.png	47	senMayoLike vs curated score scatter, females
score_tertile_correlations_adrb2_senmayo.png	48	ADRB2 by senMayoLike tertile
score_tertile_correlations_adrb2_curated.png	(sensitivity, in TSV/ figure dir only)	ADRB2 by curated tertile
score_tertile_correlations_adrb3_senmayo.png	49	ADRB3 by senMayoLike tertile
score_tertile_correlations_adrb3_curated.png	(sensitivity)	ADRB3 by curated tertile

3.2 13 new slides inserted into the deck

See §0.A of the v4.5.4 explainer for the full slide-number map. The new positions in the 50-slide deck:

- **4** — HALLMARK vs cytokine inflammation methods clarification
- **7** — ADRB3 correlation deep dive
- **28** — ADRB3 forest (broad, 50s in older)
- **29** — Sensitivity 50s-excluded (ADRB2 + ADRB3 side-by-side)
- **31** — ADRB3 Fisher-z table (3 stratifications)
- **35, 37, 39, 41** — Saltiel ADRB3 panels (4 sex × depot subsets)
- **43** — Saltiel ADRB3 summary stats
- **47** — Section 4: signature score distributions
- **48** — ADRB2 score-tertile correlations
- **49** — ADRB3 score-tertile correlations

3.3 Significance-marker convention

All v4.5.4-introduced figures use this convention consistently:

p-value	Marker
p < 0.001	***
p < 0.01	**
p < 0.05	*
p ≥ 0.05	ns (or omitted)
missing	n/a

Asterisks are placed:

- Forest plots: to the right of each 95% CI bar
- Scatter regressions: in the panel title after the gene/endpoint label
- Tabular Q1 Fisher-z slide: inline next to delta values

Caveat: existing v4.5.3 slides retain their original (pre-asterisk) appearance. Adding markers to all legacy slides was descope'd from Round 3 because those figures are not regression-critical for the meeting; the convention is established and can be retroactively applied in a future round.

4. Three top headline findings from Round 3

Finding 1 — The 50s decade may concentrate the ADRB3-fibrosis signal (v4.5.5 Slide 29; ROUND-4-HEDGED)

Older-group definition	ADRB2 × fibrosis (subq F)	ADRB3 × fibrosis (subq F)
50-79 (50s included)	$r = -0.43, p < 0.001$	$r = -0.17, \text{NS}$
60-79 (50s excluded)	$r = -0.48, p < 0.001$	$r = -0.06, \text{NS}$
50-59 only (peri window)	$r = -0.37, p < 0.01$	$r = -0.36, p < 0.01$

Round-4 Fisher-z tests for cross-stratification differences:

- ADRB3 broad-incl (-0.17) vs broad-excl (-0.06): $p = 0.43, \text{NS}$
- ADRB3 peri (-0.36) vs broad-excl (-0.06): $p = 0.054, \text{marginally sub-threshold}$

Interpretation (Round-4-hedged): ADRB2's fibrosis effect is numerically robust across age slices and intensifies when 50s are removed. For ADRB3, the directional pattern is suggestive of an effect concentrated in the 50s decade, but neither cross-stratification comparison clears the conventional $\alpha=0.05$ threshold. The "ADRB3 vanishes" framing of v4.5.4 was not supported by the Fisher-z tests. The hedged headline is: *the data are consistent with ADRB3 impairment being acute-and-transient (peaking during the perimenopausal transition) and ADRB2 impairment being lasting, but a larger cohort is needed to formally establish this.*

Finding 2 — ADRB2 vs ADRB3 specialization on the fibrosis axis is statistically robust; the other axes are directional only (v4.5.5 Slide 43; ROUND-4-HEDGED)

In female subq (n=233), Williams' tests for dependent correlations (using $r_{\text{ADRB2_ADRB3}} = +0.287$ from the same samples):

Endpoint	ADRB2 R ²	ADRB3 R ²	Williams t	p	Verdict
Age	0.021	0.047	+0.96	0.34	ns
CCL2	0.090	0.027	-1.82	0.069	ns
Fibrosis	0.174	0.026	-3.54	0.0004	***

Endpoint	ADRB2 R ²	ADRB3 R ²	Williams t	p	Verdict
Senescence	0.076	0.093	+0.41	0.68	ns

Interpretation (Round-4-hedged): only the **fibrosis** axis shows a statistically significant difference between ADRB2 and ADRB3 in female subq, with ADRB2 dominant ($p=0.0004$). The numerical Age, CCL2, and Senescence differences are directionally consistent with receptor specialization but are not statistically distinguishable at $n=233$ ($p>0.06$ for all three). The two receptors are not redundant on the fibrosis axis (formally established). The other apparent specializations are descriptive observations needing larger N. Lead manuscript framing on the fibrosis specialization; treat the others as exploratory.

Finding 3 — Score-based stratification differs from age-bin stratification (v4.5.4 Slide 47)

The senMayoLike v2 transcriptional score divides females into tertiles that **partially but imperfectly align** with age-bin pre/peri/post grouping. The percentage of "pre" samples by age that fall into the high-tertile (T3, "more post-like") is **15.4% in subcutaneous** (12 of 78 pre samples) and **29.2% in visceral** (21 of 72). Cohen's kappa between the age-bin and score-tertile classifications is **0.169 in subcutaneous and 0.100 in visceral** — both fall in the "slight agreement" range (Landis & Koch 1977: <0.20 = slight; $0.21-0.40$ = fair). The two groupings are largely discordant, supporting the use of score-based grouping as a sensitivity analysis. For the manuscript:

- **Primary stratification:** age-bin pre/peri/post (existing, conventional)
- **Sensitivity stratification:** senMayoLike score tertiles (new in v4.5.4)
- **Concordance:** report agreement and document the discordant samples

The score-based approach also enables stratification within an age band: e.g., 40-49 women can be split by score tertile to reveal whether the apparent peri-menopausal effects are uniformly distributed or concentrated in transcriptionally-advanced women within the band.

5. What was NOT changed between v4.5.3 and v4.5.4

- All 98 verified numerical claims from rounds 1+2 (no v4.5.3 numbers re-audited because v4.5.4 doesn't modify v4.5.3 prose)
 - All 23 method specifications (the new R script follows the same conventions)
 - The OVX 042126 cross-species concordance table and its labels
 - The HiRes per-cell-type DE results (Slides 17-22 = v4.5.4 Slides 19-24)
 - The Saltiel ADRB2 panels (only ADRB3 mirrors were ADDED)
 - The recommended manuscript framing (female subq Saltiel as primary; male subq as companion)
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6. Distribution policy

- **Private menopause repo** (private GitHub): receives EVERYTHING — v4.5.5 deck, explainer, review summary, and the lipolysis-paradox personal notes (per user direction; not for collaborator distribution).

- **Public menopause_open repo:** receives v4.5.5 deck, explainer, this review summary; predecessor v4.5.1/v4.5.3/v4.5.4 artifacts remain. Does **NOT** receive: the lipolysis-paradox personal notes.
- **OneDrive shared folder:** receives v4.5.5 deck, explainer, review summary. Does **NOT** receive: the lipolysis-paradox personal notes (kept private).

7. Round 4 — what was found and fixed (full ledger)

The Round-4 adversarial audit on v4.5.4 (4 reviewers, 2026-05-04) found 9 issues across factual, statistical, cross-reference, and editorial categories. All 9 are corrected in v4.5.5.

#	Severity	Where	Original (v4.5.4)	Corrected (v4.5.5)
R4-1	HIGH	Explainer §0.I	senMayoLike v2 "~30 genes with UP/DOWN"	13 genes (12 UP + 1 DOWN) — small curated subset, distinct from full published SenMayo
R4-2	HIGH	Explainer §0.I	Curated v1 "~70 genes, no direction"	72 genes used (out of 81 rows; only <code>keep=TRUE</code> retained)
R4-3	HIGH	Slide 29 + explainer §0.E	"DISAPPEARS when 50s removed... driven entirely by 50s decade"	"attenuates from -0.17 to -0.06 (Fisher-z $p=0.43$, NS); peri vs 60-79 $p=0.054$, marginal; replication needed"
R4-4	MED	Slide 43 + explainer §0.G	Specialization claimed across all 4 axes	Lead with fibrosis as only sig (Williams $t=-3.54$, $p=0.0004$); other 3 axes flagged as directional only
R4-5	MED	Slides 35, 37, 39, 41 subtitles	"Mirror of Slide 33/34/35/36 (ADRB2)"	Corrected to 34/36/38/40
R4-6	MED	Slide 43 body + explainer §0.I	"20-25% of pre in T3_high" (inaccurate)	"15.4% subq / 29.2% visc; Cohen's kappa 0.10-0.17 (slight agreement)"
R4-7	LOW	Slide 35 subtitle	" $R^2=0.094$ "	" $R^2=0.093$ " (matches TSV 0.09347)
R4-8	LOW	Slide 4 title	(cosmetic divergence)	Accepted as-is; no fix
R4-9	LOW	Review summary §3.1	"sensitivity_50s_excluded_fisherz.tsv: 17 rows"	Corrected to 68 rows
R4-10	LOW	Lipolysis personal notes	ALK7 → C/EBPα → PNPLA2/PLIN1 chain presented as established	Hedged: Smad2/3 alternative noted; C/EBPα intermediate flagged as inference

Round-4 footers: all 50 slides updated v4.5.4 | <#> → v4.5.5 | <#> .

Net status after Round 4:

- **0 numerical errors** in any deck or explainer claim
- **0 R-script methods errors** (R logic fully verified clean by R12)
- **0 remaining factual or interpretation overstatements**

- v4.5.5 is the version recommended for collaborator distribution.
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8. Open follow-ups for a hypothetical Round 5

1. Apply the asterisk significance-marker convention to legacy v4.5.3 figures (currently only v4.5.4-new figures carry the markers).
2. Validate score-tertile vs age-bin grouping by computing effect sizes within each scheme on a key endpoint (e.g., ADRB2-fibrosis correlation) and comparing — this is the "outcome validation" step that would let the explainer recommend score-based as primary rather than sensitivity.
3. Replicate the ADRB3-fibrosis "50s concentration" pattern in an independent cohort to push the Fisher-z test $p=0.054$ across the conventional threshold.
4. Cell-type-resolved ADRB2 + ADRB3 trajectories using existing CIBERSORTx HiRes outputs (the per-cell-type expression matrices on disk; Slide 50 next-steps item).
5. Add PDE3B, AKT2, G0S2, ADORA1/2A, GRK2/ARRB2/RGS2 trajectories to the deck — the lipolysis-paradox personal notes argue these are the missing pieces of the receptor-desensitization circuit.
6. Cell-type-resolved ADRB2 + ADRB3 trajectories using existing CIBERSORTx HiRes outputs (the per-cell-type expression matrices on disk; Slide 50 next-steps item).
7. Add PDE3B, AKT2, G0S2, ADORA1/2A, GRK2/ARRB2/RGS2 trajectories to the deck — the lipolysis-paradox personal notes argue these are the missing pieces of the receptor-desensitization circuit.
8. Verify that the `concordance_with_ovx_042126.tsv` ADRB3 rows (CONCORDANT for both depots) hold up if computed using only the score-tertile-T3 ("post-like") subset of females rather than the full older age bin.